

7 - Assignment on theoretical data transformations

Problem 1

Movies rated	Mov 1	Mov 2	Mov 3	Movie 4	Movie 5	Movie 6	Movie 7	Movie 8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

Use collaborative filtering to predict User 2's preference for Movie 6. The data frame is shown below:

```
1 df <- data.frame(  
2   x = c(5, 1, 4, NA, NA, NA, 2, 1),  
3   y = c(5, NA, 3, 2, 5, NA, 2, 1),  
4   z = c(1, 4, 2, 5, 2, NA, 5, 4),  
5   t = c(4, 2, NA, 1, 5, 2, 3, 3)  
6 )
```

Problem 2

Use Pearson correlation to determine which movie is the most similar to Movie 1.

(Optional: Google how to use a for loop to speed up this task.)

Problem 3A

Apply mean imputation grouped according to region to fill in missing values in the column for "Agriculture" and the column for "Industry" for the countries dataset from

<http://manctsui.github.io/datasets/countries.csv>

Problem 3B

How many NA's did you have to fill from each region?

Problem 4 (theoretical)

The data frame below contains data of outdoor temperature vs. the number of times a cricket chirps within 15 seconds:

```
1 df <- data.frame(  
2   temp = c(79, 74, 73, 76, 82, 72, 71, 79, 83, 70, 73, 75, 77),  
3   chirps = c(44, 40, 38, 43, 48, 37, 34, 45, 51, 32, 35, 42, 43)  
4 )
```

Running `lm(temp ~ chirps, data = df)` shows that $temp = 0.714 \times chirp + 46.474$ is the line of best fit relating temperature with cricket chirps:

Call:

```
lm(formula = temp ~ chirps, data = df)
```

Coefficients:

(Intercept)	chirps
46.474	0.714

Explain via a manual computation, how finding `cor(df$temp, df$chirps)`, `mean(df$temp)`, `sd(df$temp)`, `mean(df$chirps)`, `sd(df$chirps)` results in the above linear regression equation as well.